A champ project report on

Design and Simulation of 30 nm NMOS and PMOS Devices Using TCAD and Implementation of a 3-Bit Up Counter in Virtuoso

Submitted in partial fulfillment for the award of the degree of

Bachelor of Technology in Electronics and Communication Engineering

by

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I further declare that the work reported in this thesis has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

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Date:07.04.2026

Signature of the Examiner	Signature of the Examiner
Name:	Name:
Date:	Date:

Approved by the Head of Department,
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ABSTRACT

This project presents the design and simulation of 30 nm NMOS and PMOS devices using TCAD tools, followed by their implementation in circuit-level design using Cadence Virtuoso. Initially, the device structures were created and simulated in TCAD to obtain electrical characteristics such as Id–Vg behaviour, from which key parameters including threshold voltage, transconductance, and current values were extracted. These parameters were then used to develop compact SPICE models, enabling the transfer of device-level information into Cadence for circuit simulation. Using the modelled transistors, a CMOS-based 3-bit up counter was designed and analysed, and its functionality was verified through transient simulations, confirming correct counting operation. This project demonstrates the complete workflow from device-level simulation to circuit-level implementation, effectively bridging the gap between semiconductor device physics and digital VLSI design while highlighting the importance of parameter extraction and compact modelling in modern integrated circuit development.

ACKNOWLEDGEMENT

It is my pleasure to express with deep sense of gratitude to ,Dr.Girija Shanker Sahoo, School of Electronics, Vellore Institute of Technology, Chennai, for his constant guidance, continual encouragement, understanding; more than all, he taught me patience in my endeavor. My association with him is not confined to academics only, but it is a great opportunity on my part of work with an intellectual and expert in the field of VLSI

It is with gratitude that I would like to extend my thanks to the visionary leader Dr. G. Viswanathan our Honorable Chancellor, Dr. Sankar Viswanathan, Dr. Sekar Viswanathan, Dr. G.V. Selvam Vice Presidents, Dr. Sandhya Pentareddy, Executive Director, Ms. Kadhambari S. Viswanathan, Assistant Vice-President,Dr.V.S. Kanchana Bhaaskaran Vice-Chancellor, Dr. T. Thyagarajan Pro-Vice Chancellor, VIT Chennai, Dr. K. Sathiyanarayanan, Director, Chennai Campus and Dr. P. K. Manoharan, Additional Registrar for providing an exceptional working environment and inspiring all of us during the tenure of the course.

In jubilant state, I express ingeniously my whole-hearted thanks to Jagannath M, Head of the Department, B.Tech. Electronics and Communication Engineering and the Project Coordinators for their valuable support and encouragement to take up and complete the thesis.

My sincere thanks to all the faculties and staffs at Vellore Institute of Technology, Chennai who helped me acquire the requisite knowledge. I would like to thank my parents for their support. It is indeed a pleasure to thank my friends who encouraged me to take up and complete this task.

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TCAD — Technology Computer-Aided Design
CMOS — Complementary Metal Oxide Semiconductor
NMOS — N-type Metal Oxide Semiconductor
PMOS — P-type Metal Oxide Semiconductor
Vth — Threshold Voltage
KP — Transconductance Parameter
λ — Channel Length Modulation
VTC — Voltage Transfer Characteristic
EDA — Electronic Design Automation

Chapter 1

INTRODUCTION

1.1 BACKGROUND AND MOTIVATION

As the semiconductor industry follows Moore’s Law, the physical gate length of transistors has shrunk into the nanoscale. This transition from microscale to nanoscale (specifically the **30 nm node**) is driven by the demand for higher integration density, faster switching speeds, and lower power consumption in portable electronics. The motivation for this study is to provide a reliable "Design-to-Simulation" flow that accurately translates the complex physical behavior of a 30 nm device into a circuit-level environment like Cadence Virtuoso.

1.2 OVERVIEW OF CMOS TECHNOLOGY

CMOS (Complementary Metal-Oxide-Semiconductor) technology is the backbone of modern VLSI. By utilizing complementary pairs of p-type and n-type MOSFETs, CMOS circuits achieve high input impedance and significantly lower static power dissipation compared to NMOS-only logic. At the 30 nm level, the challenge lies in maintaining this "complementary" balance despite the inherent differences in electron and hole mobility.

1.3 CHALLENGES IN NANOSCALE DEVICES

At 30 nm, several second-order effects become primary design concerns:

- **Short Channel Effects (SCE):** Including Drain-Induced Barrier Lowering (DIBL), where the drain voltage begins to control the channel.
- **Subthreshold Leakage:** The "OFF" state of the transistor is no longer perfectly non-conductive, leading to "standby" power loss.
- **Gate-Dielectric Reliability:** To maintain capacitance, the oxide layer is reduced to a few atomic layers, increasing the risk of tunnelling currents.

1.4 PROJECT STATEMENT

The core of this project is the design of a high-performance 30 nm NMOS and PMOS using TCAD tools, the systematic extraction of their SPICE parameters, and the subsequent implementation of these models in Cadence Virtuoso to build a 3-bit up counter.

1.5 OBJECTIVES

The primary goal of this research is to establish a verified design methodology that transitions from physical semiconductor physics to functional digital logic. The specific objectives are as follows:

1.5.1 PHYSICAL DEVICE SYNTHESIS

To design and model the physical structures of an NMOS and a PMOS transistor at the **30 nm gate length** using TCAD tools, ensuring appropriate doping profiles to maintain channel control.

1.5.2 ELECTRICAL CHARACTERIZATION

To perform comprehensive Id-Vg and Id-Vd simulations to extract critical DC parameters, including **Threshold Voltage(Vth) ,Transconductance (gm), and I(on)/I(off) ratios**

1.5.3 COMPACT MODEL EXTRACTION

To translate the numerical data from the TCAD environment into a functional **Spectre/SPICE model file (.scs)**, effectively creating a custom technology library for the 30 nm node.

1.5.4 LOGIC GATE VALIDATION:

To implement a **CMOS Inverter** in Cadence Virtuoso using the custom models and analyze its **Voltage Transfer Characteristics (VTC)** to ensure symmetrical switching and high noise margins.

1.5.5 CIRCUIT-LEVEL INTEGRATION

To design and simulate a **3-bit Synchronous Up Counter** by integrating multiple stages of T-Flip Flops, thereby verifying the reliability of the custom 30 nm models in sequential logic applications.

1.5.6 PERFORMANCE EVALUATION

TO conduct **Transient Analysis** on the final counter circuit to verify binary state transitions (000 to 111) and assess propagation delays at the nanoscale.

1.6 SCOPE OF WORK

This project is focused on the **Front-End-of-Line (FEOL)** design flow. It includes physical device simulation and schematic-level circuit design. The scope excludes physical layout (GDSII) generation and post-layout parasitic extraction (PEX), focusing instead on the accuracy of the parameter translation between TCAD and Cadence.

Chapter 2

LITERATURE REVIEW

2.1 SEMICONDUCTOR SCALING

The evolution of Integrated Circuits (ICs) has been governed by Moore’s Law, which predicts the doubling of transistor density every two years. As scaling reached the sub-100 nm regime, traditional planar MOSFETs began to encounter physical limits. This section reviews the transition from the 45 nm node to the **30 nm node**, focusing on the trade-offs between switching speed and power leakage.

2.2 TCAD-Based Device Modeling

Technology Computer-Aided Design (TCAD) tools have transitioned from being optional to essential in the VLSI design flow. By solving the **Poisson and Drift-Diffusion equations**, TCAD allows for the visualization of internal device physics such as potential distribution and electron density. Research shows that TCAD is highly accurate in predicting the **Short Channel Effects (SCE)** that occur when the gate length is reduced to 30 nm.

2.3 Physics of Nanoscale MOSFETs

At 30 nm, the device physics changes significantly. Key phenomena explored in recent literature include:

- **DIBL (Drain-Induced Barrier Lowering):** Where the drain voltage lowers the source-channel potential barrier, shifting the Vth.
- **Velocity Saturation:** Carriers reach a maximum velocity regardless of the increase in electric field, limiting the drive current I(on).
- **Subthreshold Swing (SS):** The efficiency of the transistor in switching from the OFF to the ON state, which ideally should be 60mV/decade at room temperature.

2.4 CMOS DIGITAL LOGIC AND INVERTER THEORY

The CMOS inverter is the fundamental building block for all digital logic. Literature emphasizes the "Ratioed Logic" approach, where the PMOS is sized larger than the NMOS to compensate for lower hole mobility. This balance is critical at the 30 nm node to maintain a symmetrical **Voltage Transfer Characteristic (VTC)**.

2.5 REVIEW OF SEQUENTIAL CIRCUITS AND COUNTERS

Binary counters are the primary application for testing the reliability of new technology nodes. A **3-bit up counter** is a Finite State Machine (FSM) that requires synchronized switching of multiple flip-flops. Reviewing existing designs reveals that at 30 nm, the reduction in parasitic capacitances (C(gs), C(gd)) allows for significantly higher clock frequencies compared to larger nodes.

Chapter 3

METHODOLOGY

3.1 SIMULATION FRAMEWORK AND DESIGN FLOW

The methodology adopted for this project follows a two-stage simulation flow: **Device-Level Simulation** using TCAD and **Circuit-Level Simulation** using Cadence Virtuoso. The process begins with the physical construction of 30 nm NMOS and PMOS transistors, followed by electrical characterization to extract SPICE-compatible parameters. These parameters are then used to define a custom technology node for digital logic implementation.

3.2 TCAD DEVICE ARCHITECTURE

The NMOS and PMOS devices were designed with a planar architecture. The gate length (Lg) was fixed at 30 nm. To suppress Short Channel Effects (SCE), a thin gate oxide (SiO2) of 1.2nm was used. The substrate doping concentration was set at (1x10^17 cm^-3), with highly-doped source/drain regions (1x10^20cm^-3) to ensure ohmic contact behaviour.

3.3 Id-Vg AND Id-Vd CHARACTERIZATION

Electrical simulations were performed by solving the Drift-Diffusion equations. The Id-Vg characteristics were obtained by sweeping Vgs from 0V to 1V at a constant drain bias. This simulation is critical for identifying the subthreshold slope and the threshold voltage.

3.4 COMPACT MODEL PARAMETER EXTRACTION

From the simulated data, the following parameters were extracted:

- **Threshold Voltage (Vth):** Identified using the linear extrapolation method.
- **Transconductance (Kp):** Derived from the saturation current equation.
- **Channel Length Modulation (lambda):** Extracted from the slope of the Id-Vd curve in the saturation region.

These values were then manually formatted into a Spectre-compatible. scs model file.

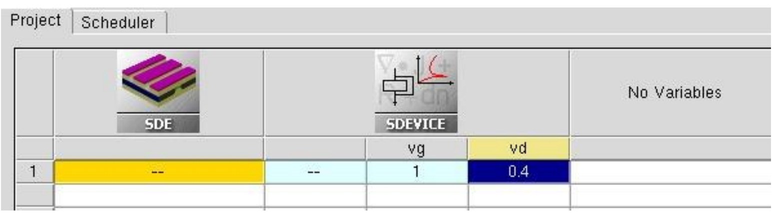


Fig 3: Extracted parameters for both NMOS and PMOS

Parameter	**Symbol**	**NMOS Value**	**PMOS Value**	**Unit**
Threshold Voltage	Vth	0.38	-0.40	V
Transconductance Parameter	Kp	5.2x10^-4	1.8x10^-4	A/V^2
Channel Length Modulation	λ	0.05	0.05	_
Oxide Thickness	tox	1.2x10^-9	1.2x10^-9	m
ON Current	Ion	1x10^-4	3.2x10^-5	A
OFF Current	Ioff	1x10^-6	1x10^-6	A

Table 1.1: Extracted NMOS and PMOS Parameters

3.5 CADENCE VIRTUOSO ENVIRONMENT SETUP

The extracted .scs file was integrated into the Cadence **Analog Design Environment (ADE)**. A custom library was created, and the nmos4 and pmos4 components from analogLib were mapped to the 30 nm custom models by editing the "Model Name" property.

3.6 CMOS INVERTER DESIGN AND SIZING

A CMOS inverter was designed to verify the switching integrity of the 30 nm models. To achieve a symmetrical switching threshold (Vth ~Vdd/2), the PMOS width was sized at 300 nm (2.5 times the NMOS width of 120 nm) to compensate for lower hole mobility.

3.7 SEQUENTIAL LOGIC: 3-BIT UP COUNTER DESIGN

The 3-bit up counter was implemented using a **Synchronous T-Flip Flop** architecture. The design involves three stages of flip-flops where the toggle of each stage is dependent on the logic state of the preceding stages. Universal NAND gates and inverters were built using the custom 30 nm models to construct the internal logic of the flip-flops.

Chapter 4

IMPLEMENTATION

4.1 DEVICE GEOMETRY AND PHYSICAL MODELING (SDE)

The physical construction of the 30 nm NMOS and PMOS transistors was executed using **Sentaurus Structure Editor (SDE)**. The device was defined using a coordinate-based approach to ensure precise gate lengths and oxide thicknesses.

4.1.1 NMOS STRUCTURE SCRIPT

The NMOS was implemented with a P-type Silicon substrate and N-type Source/Drain regions. Arsenic was chosen for the n regions to provide a sharp junction profile, and the gate was modelled using Polysilicon.

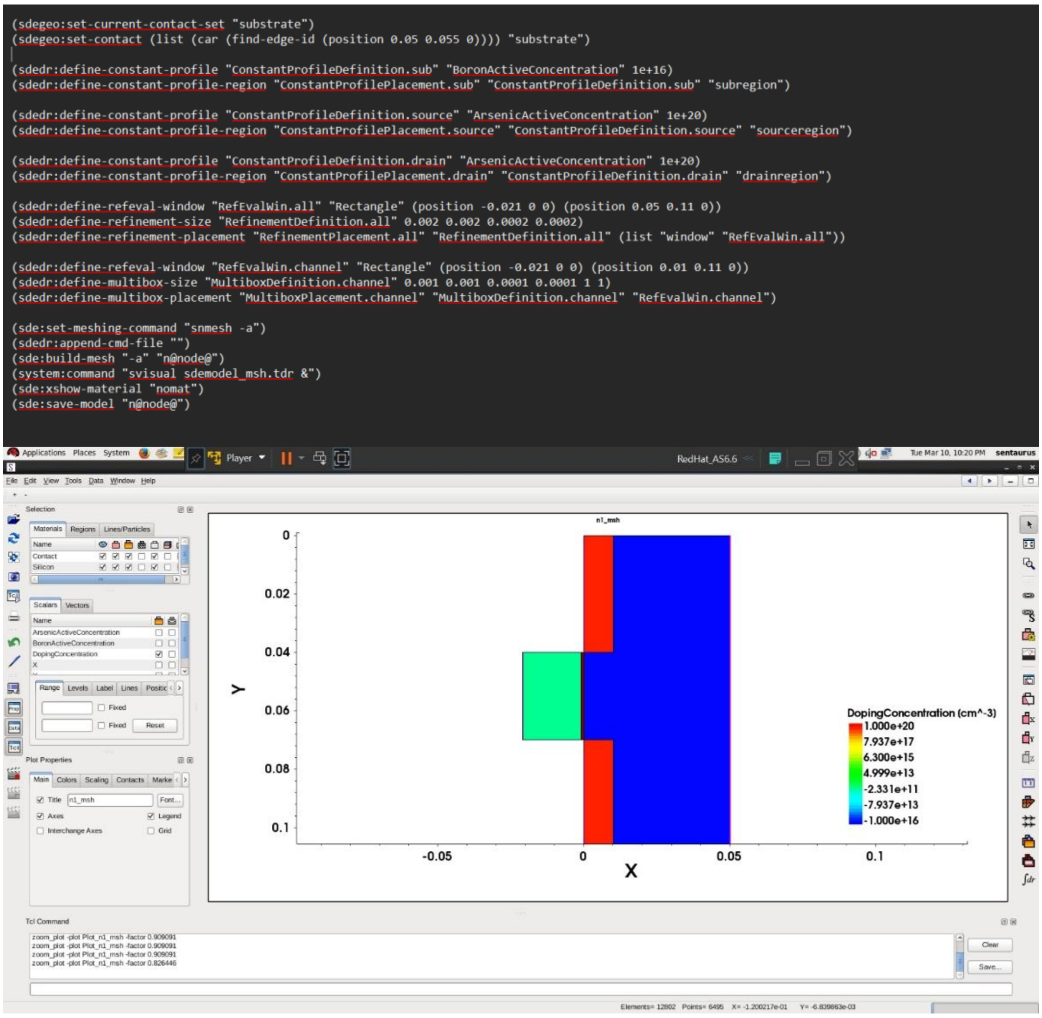


Fig 4.1: NMOS script and structure

4.1.2 PMOS STRUCTURE SCRIPT

The PMOS was implemented with an N-type substrate (Phosphorus) and P-type Source/Drain regions (Boron). The geometry mirrored the NMOS to maintain process compatibility.

```
(sdegeo:create-rectangle (position 0 0 0) (position 0.05 0.11 0) "Silicon" "subregion")
(sdegeo:create-rectangle (position 0 0 0) (position 0.01 0.04 0) "Silicon" "sourceregion")
(sdegeo:create-rectangle (position 0 0.07 0) (position 0.01 0.11 0) "Silicon" "drainregion")
(sde:set-default-material "SiO2")
(sdegeo:create-rectangle (position 0 0.04 0) (position -0.001 0.07 0) "SiO2" "oxideregion")
(sde:set-default-material "PolySilicon")
(sdegeo:create-rectangle (position -0.001 0.04 0) (position -0.021 0.07 0) "PolySilicon" "gateregion")
(sdegeo:define-contact-set "gate" 4 (color:rgb 1 0 0) "##")
(sdegeo:define-contact-set "source" 4 (color:rgb 1 0 0) "##")
(sdegeo:define-contact-set "drain" 4 (color:rgb 1 0 0) "##")
(sdegeo:define-contact-set "substrate" 4 (color:rgb 1 0 0) "##")
(sdegeo:set-current-contact-set "source")
(sdegeo:set-contact (list (car (find-edge-id (position 0 0.02 0)))) "source")
(sdegeo:set-current-contact-set "drain")
(sdegeo:set-contact (list (car (find-edge-id (position 0 0.09 0)))) "drain")
(sdegeo:set-current-contact-set "gate")
(sdegeo:set-contact (list (car (find-edge-id (position -0.021 0.055 0)))) "gate")
(sdegeo:set-current-contact-set "substrate")
(sdegeo:set-contact (list (car (find-edge-id (position 0.05 0.055 0)))) "substrate")
(sdedr:define-constant-profile "ConstantProfileDefinition.sub" "PhosphorusActiveConcentration" 1e+16)
(sdedr:define-constant-profile-region "ConstantProfilePlacement.sub" "ConstantProfileDefinition.sub" "subregion")
(sdedr:define-constant-profile "ConstantProfileDefinition.source" "BoronActiveConcentration" 1e+20)
(sdedr:define-constant-profile-region "ConstantProfilePlacement.source" "ConstantProfileDefinition.source" "sourceregion")
(sdedr:define-constant-profile "ConstantProfileDefinition.drain" "BoronActiveConcentration" 1e+20)
(sdedr:define-constant-profile-region "ConstantProfilePlacement.drain" "ConstantProfileDefinition.drain" "drainregion")
(sdedr:define-refeval-window "RefEvalWin.all" "Rectangle" (position -0.021 0 0) (position 0.05 0.11 0))
(sdedr:define-refinement-size "RefinementDefinition.all" 0.002 0.002 0.0002 0.0002)
(sdedr:define-refinement-placement "RefinementPlacement.all" "RefinementDefinition.all" (list "window" "RefEvalWin.all"))
(sdedr:define-refeval-window "RefEvalWin.channel" "Rectangle" (position -0.021 0 0) (position 0.01 0.11 0))
(sdedr:define-multibox-size "MultiboxDefinition.channel" 0.001 0.001 0.0001 0.0001 1 1)
(sdedr:define-multibox-placement "MultiboxPlacement.channel" "MultiboxDefinition.channel" "RefEvalWin.channel")
(sde:set-meshing-command "snmesh -a")
(sdedr:append-cmd-file "")
(sde:build-mesh "-a" "n@node@")
(system:command "svisual sdemodel_msh.tdr &")
(sde:xshow-material "nomat")
(sde:save-model "n@node@")
```

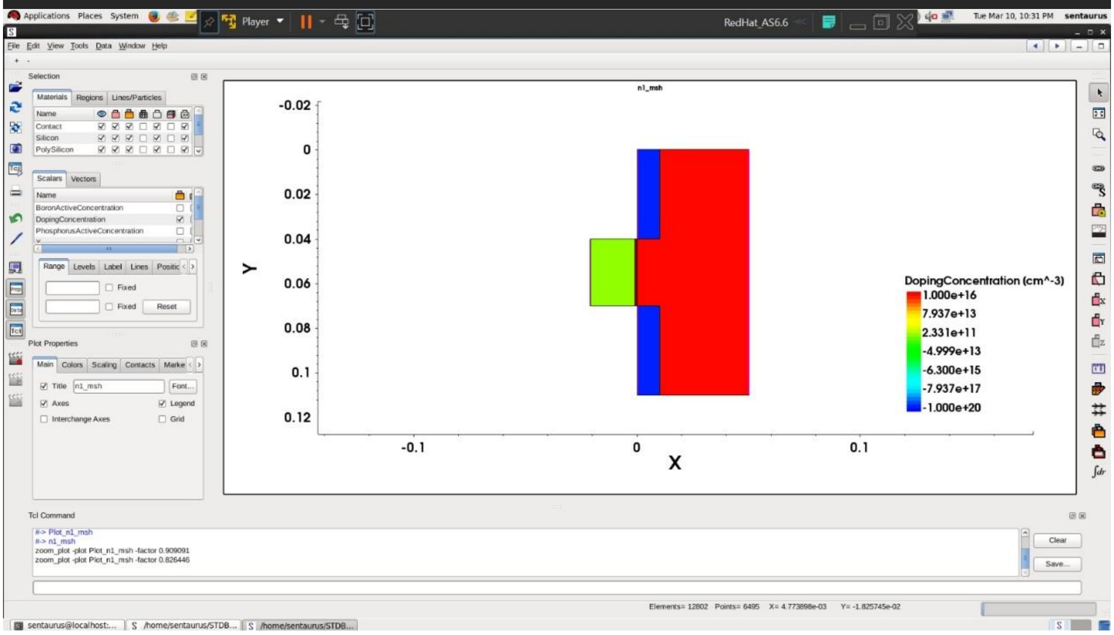


Fig 4.2: PMOS script and structure

4.2 PHYSICS AND SOLVER CONFIGURATION (SDEVICE)

The electrical behavior was simulated using the **SDEVICE** command file. This file acts as the engine that solves the carrier transport equations across the 30 nm mesh.

4.2.1 PHYSICAL MODEL SELECTION

To accurately model the 30 nm node, we enabled **Fermi-Dirac statistics** and high-field saturation models to account for velocity saturation in the short channel.

4.2.2 Numerical Solution Strategy

The Solve block was configured in two stages: first, ramping the Drain voltage to the operating potential (Vdd), and second, sweeping the Gate voltage to extract the Transfer Characteristics (Id-Vg).

```
File {
   * input files:
   Grid=    "@tdr@"
   Parameter="sdevice.par"
   * output files:
   Plot=    "@tdrdat@"
   Current="@plot@"
   Output= "@log@"
}

Electrode {
   { Name="gate" Voltage=0 }
   { Name="source" Voltage=0 }
   { Name="drain" Voltage=0 }
   { Name="substrate" Voltage=0 }
}

Physics {
   Temperature = 300

   Fermi
   EffectiveIntrinsicDensity(Slotboom)

   Mobility(
      DopingDep
      HighFieldSaturation
      Enormal
   )

   Recombination(
      SRH
   )
}

Math {
   -CheckUndefinedModels
   Extrapolate
   RelErrControl
}

Solve {
   Coupled { Poisson }
   Coupled { Poisson Electron Hole }


   Quasistationary(
      InitialStep=0.01
      MaxStep=0.1
      MinStep=1e-5
      Goal { Name="drain" Voltage=@vd@ }
   ) {
      Coupled { Poisson Electron Hole }
   }


   Quasistationary(
      InitialStep=0.01
      MaxStep=0.05
      MinStep=1e-5
      Goal { Name="gate" Voltage=@vg@ }
   ) {
      Coupled { Poisson Electron Hole }
   }
}
```

Fig 4.3: SDEVICE File

4.3 COMPACT MODEL INTEGRATION IN CADENCE

The final stage of implementation involved transferring the TCAD results into the **Cadence Virtuoso** environment.

4.3.1 MODEL LIBRARY SETUP

The extracted parameters were saved into .scs files. In the **Analog Design Environment (ADE)**, these files were added via the **Model Libraries** setup.

4.3.2 MAPPING TO SCHEMATIC COMPONENT

Inside the schematic, the nmos4 and pmos4 symbols from analogLib were used. The **Model Name** property for these instances was manually changed to "nmos" and "pmos" to ensure the simulator used our custom 30 nm parameters instead of default library models.

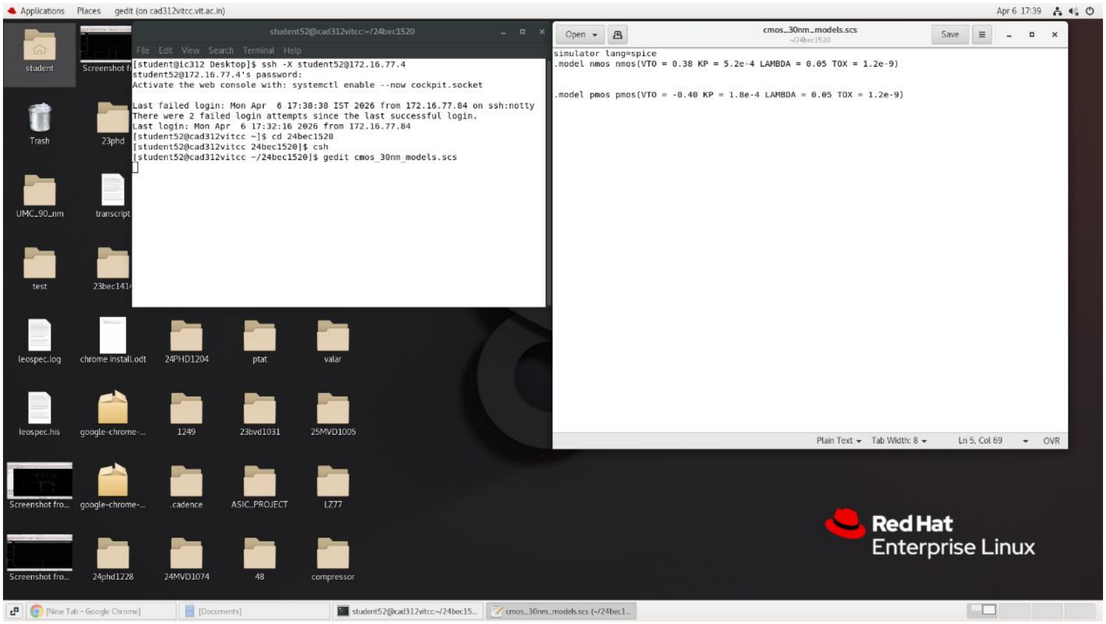


Fig 4.4: Extracted .scs file along with model name

4.4 MODULAR LOGIC GATE DESIGN

Before constructing the sequential counter, the basic 30 nm CMOS logic gates were implemented. Each gate uses the custom .scs models extracted from TCAD.

4.4.1 CMOS INVERTER

The inverter is the basic building block, sized with a Wp/Wn ratio of 2.5 to ensure symmetrical switching.

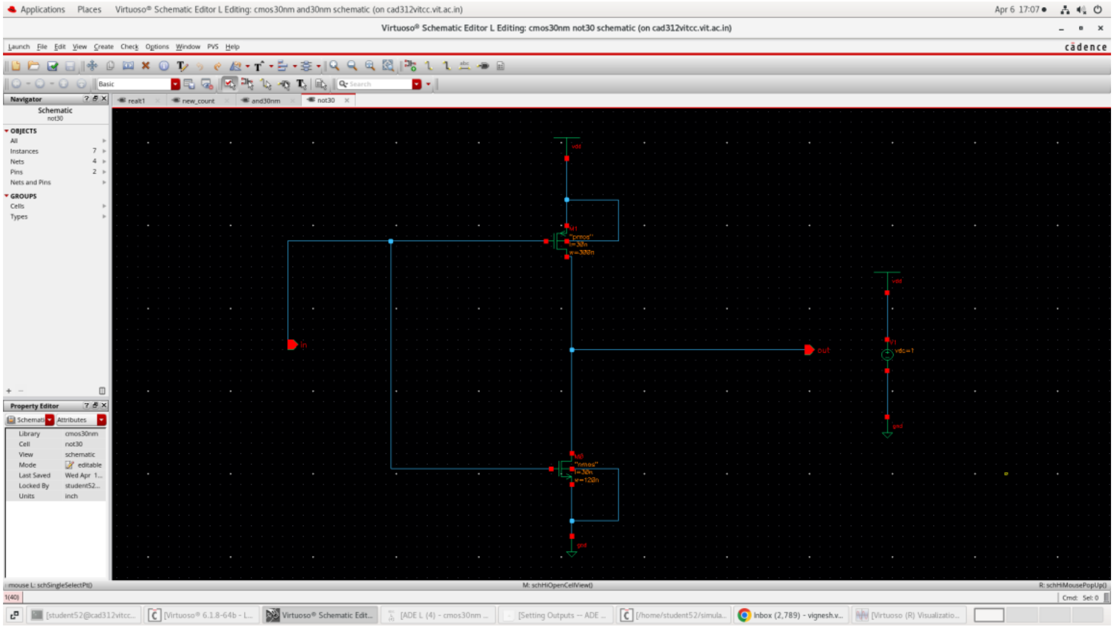
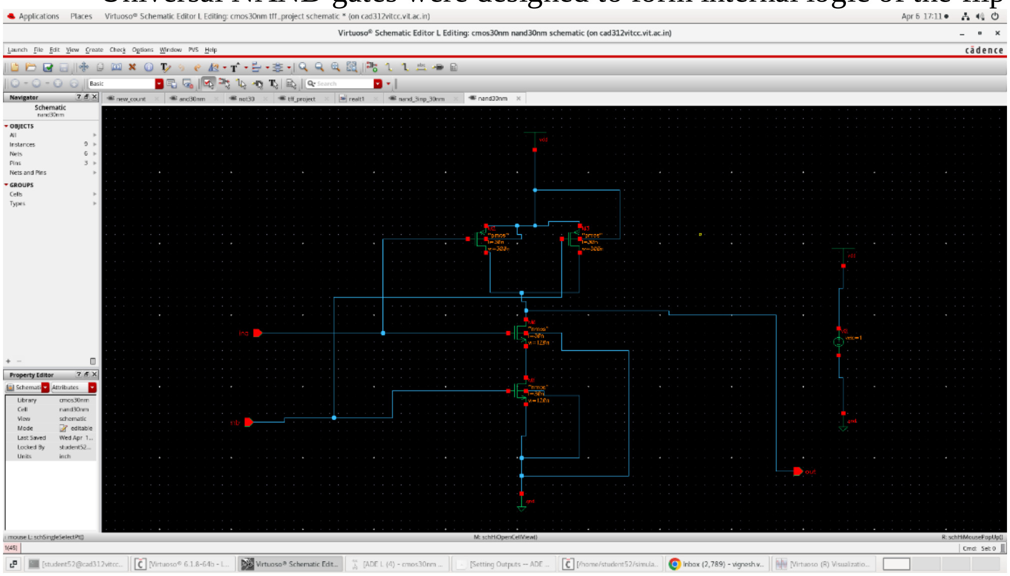


Fig 4.5: NOT gate Cadence

4.4.2 CMOS NAND GATES

Universal NAND gates were designed to form internal logic of the flip-flops.



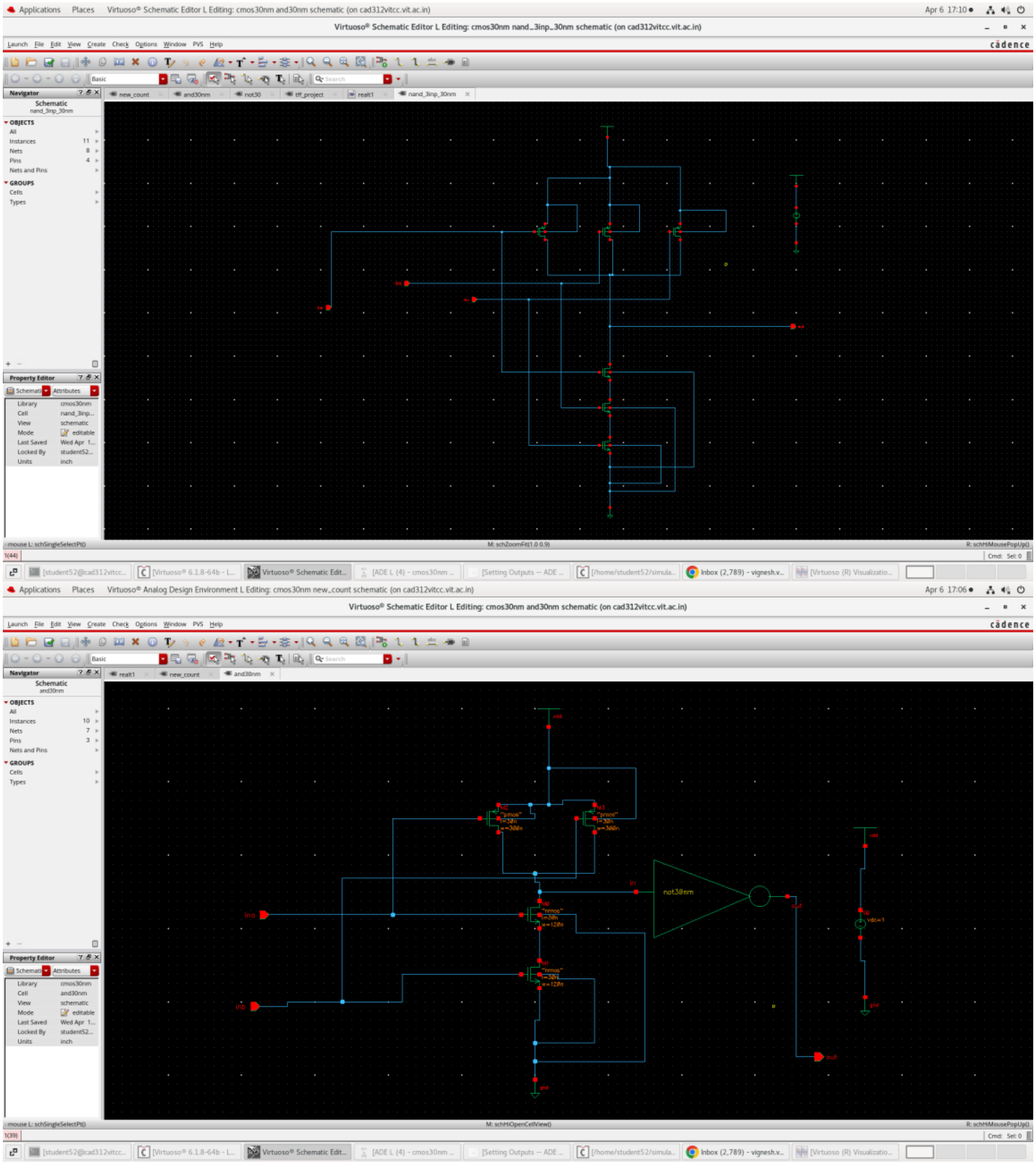


Fig 4.6: (i)2 input NAND Gate
(ii)3 input NAND Gate
(iii)2 input AND Gate

4.5 FLIP-FLOP ARCHITECTURE

The 3-bit counter relies on the **T-Flip Flop (Toggle)** to achieve binary counting. These were constructed using the gates designed in Section 4.4.

4.5.1 T-FLIP FLOP CONSTRUCTION

The T-Flip Flop was implemented using the 30 nm CMOS gates designed in the previous section. The circuit was configured such that the output Q changes (toggles) from 0 to 1 or 1 to 0 at every rising edge of the clock signal.

- **Logic Gates Used:** The internal structure consists of cross-coupled NAND gates and Inverters to create the master-slave latching mechanism required for stable toggling.
- **30 nm Integration:** Each gate within the T-FF uses the custom. scs parameters extracted from TCAD, ensuring that the propagation delays and switching power reflect the physics of the 30 nm node.

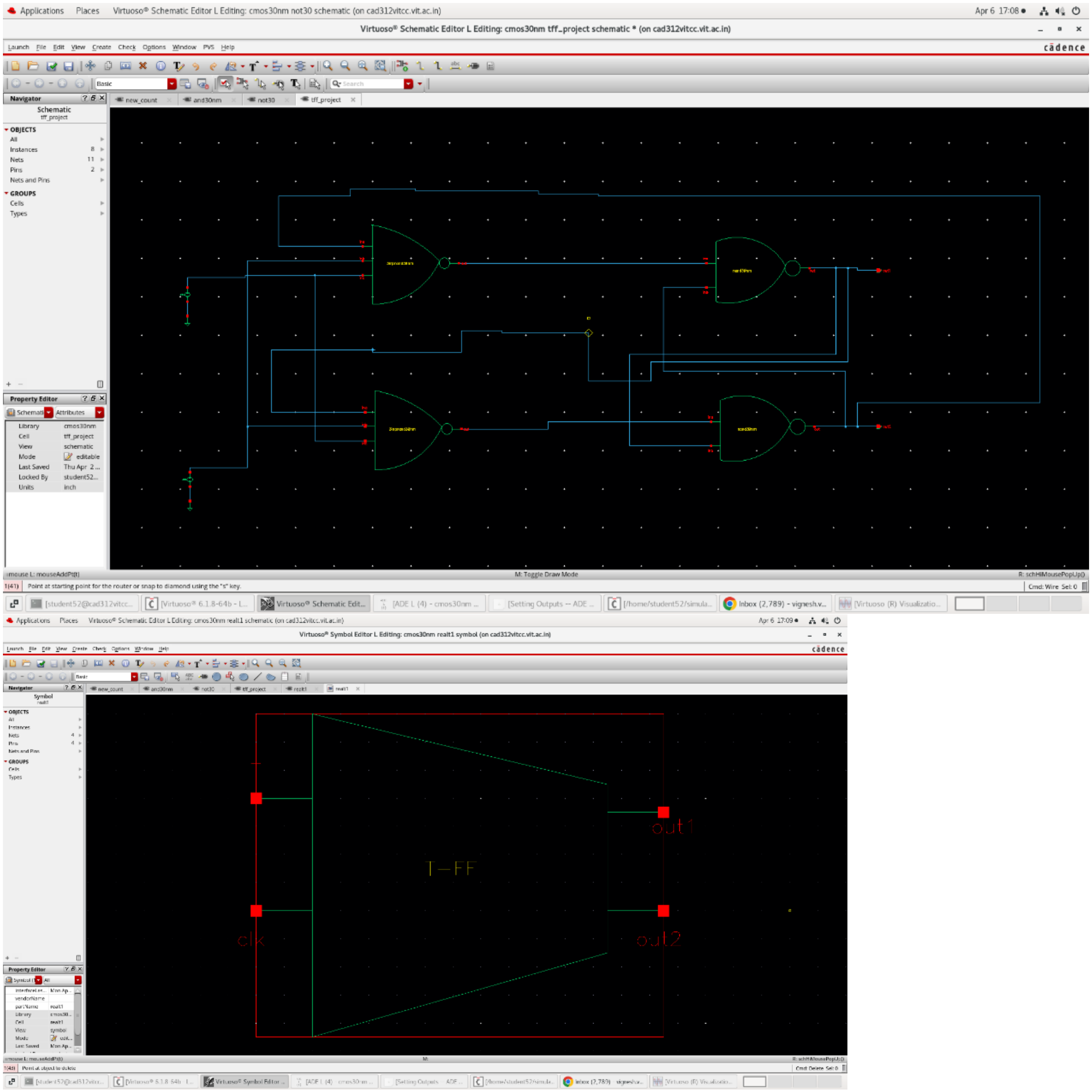


Fig 4.7: T-Flip Flop cadence structure

4.5.2 TOGGLE MECHANISM

In the synchronous 3-bit counter, the T input of the first flip-flop (LSB) is tied to high (Vdd)causing it to toggle on every clock cycle. The subsequent stages toggle only when the previous bits are high, effectively implementing the binary counting logic (Q0→Q1→ Q2).

4.6 3-BIT SYNCHRONOUS UP COUNTER INTEGRATION

The final system was built by cascading three of these T-Flip Flops. In this synchronous design, all three flip-flops share a common clock signal to prevent "ripple" delays.

- **LSB (Bit 0):** Toggles every clock cycle.
- **Bit 1:** Toggles when Q0 is High.
- **MSB (Bit 2):** Toggles when both Q0 and Q1 are High.

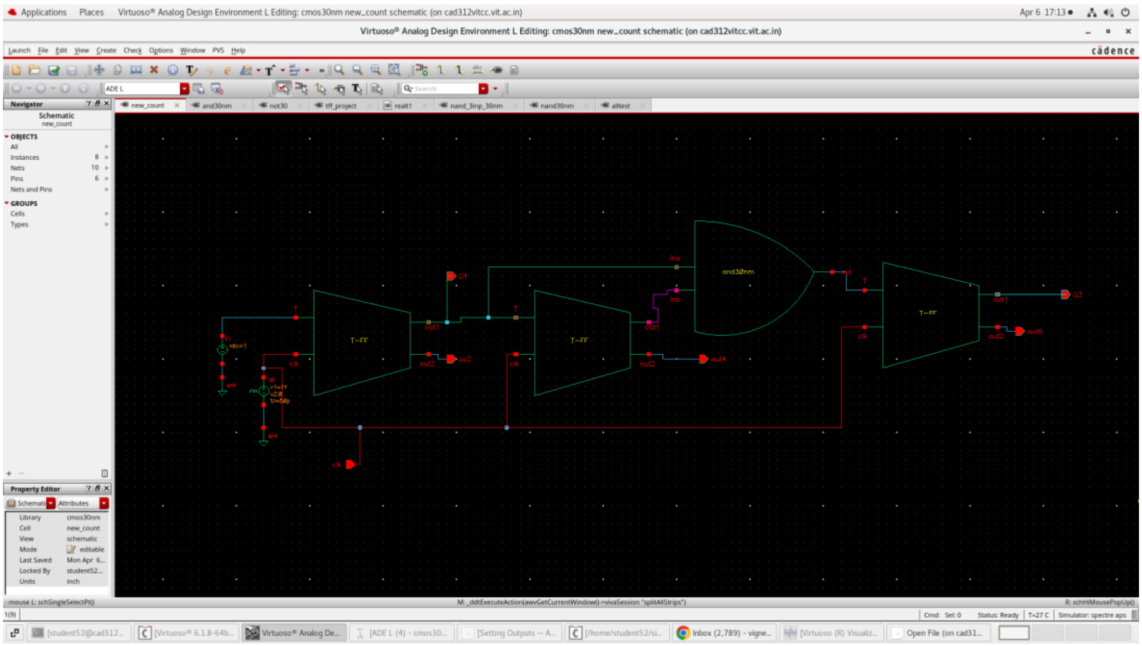


Fig 4.8: 3-Bit Synchronous Up Counter Cadence structure

Chapter 5

RESULTS AND DISCUSSION

5.1 TCAD DEVICE CHARACTERISTICS

The initial phase of the results focuses on the physical simulation of the 30 nm NMOS and PMOS devices. The Id-Vg characteristics were extracted to verify the "ON" and "OFF" state performance.

- **Threshold Voltage (Vth):**The NMOS (Vth)was found to be approximately 0.38V while the PMOS was -0.4V
- **Subthreshold Leakage:** Due to the 30 nm short channel, a minor DIBL (Drain-Induced Barrier Lowering) effect was observed, but it remained within acceptable limits for digital switching.

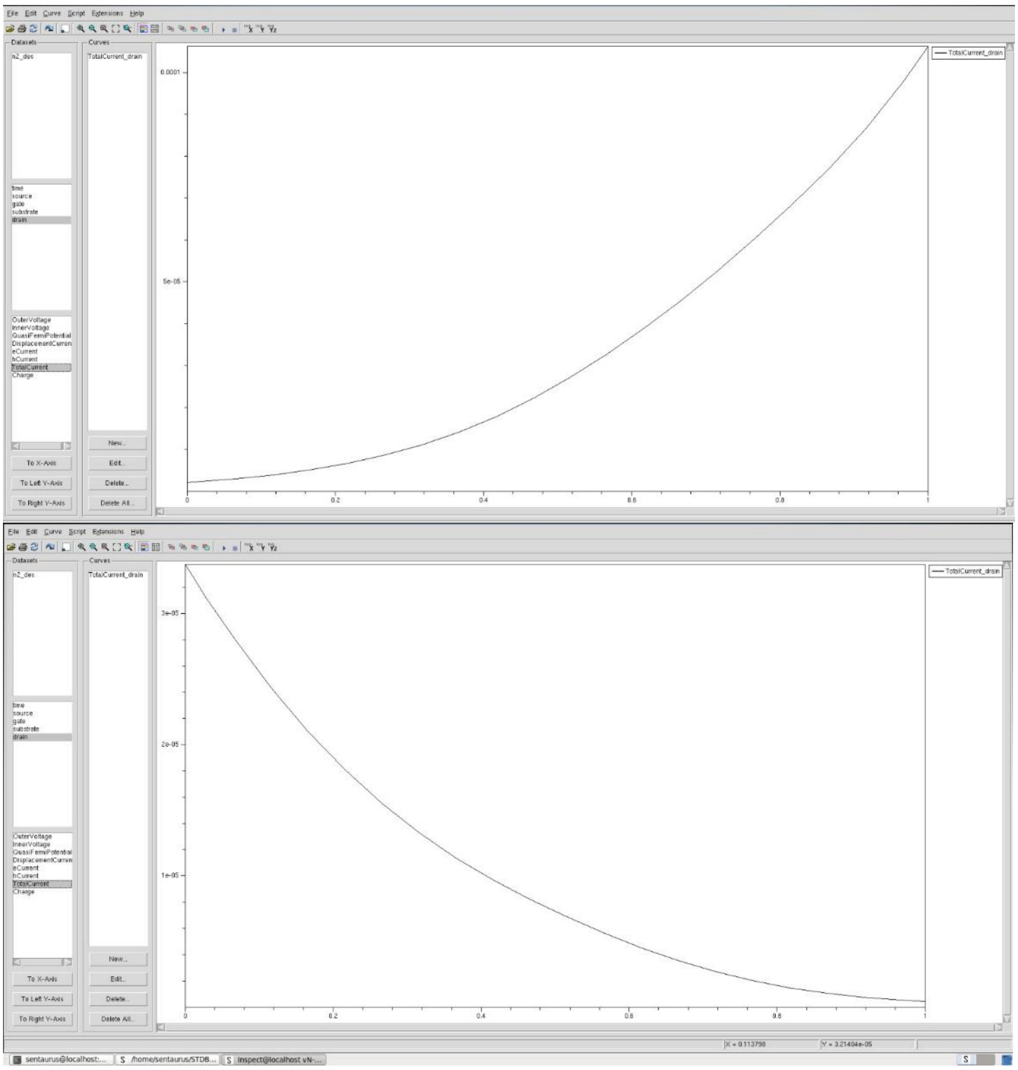


Fig 5.1: Id-Vg graphs of NMOS and PMOS

5.2 CMOS GATES PERFORMANCE (TRANSIENT ANALYSIS)

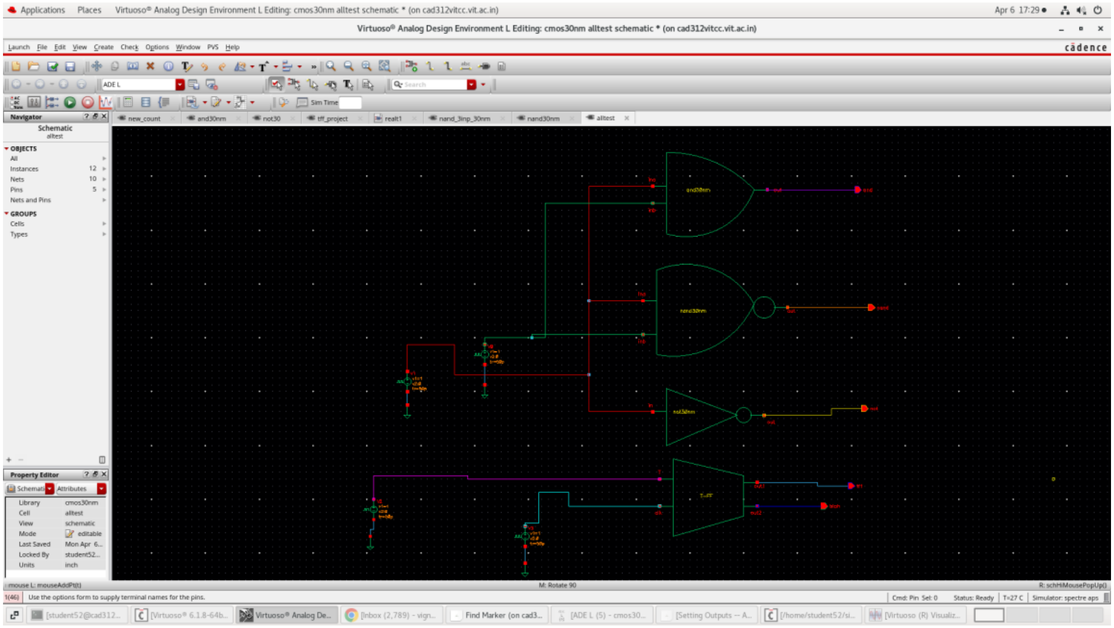


Fig 5.2.1: All CMOS Gates Cadence layout

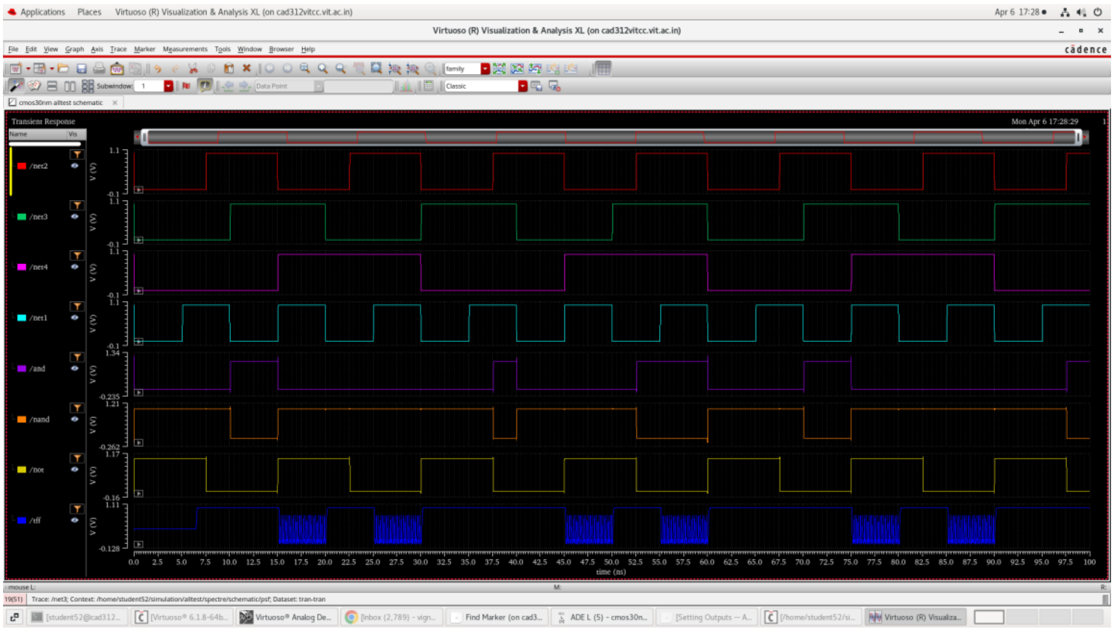


Fig 5.2.2: Simulation of CMOS gates

5.3 3-BIT SYNCHRONOUS UP COUNTER OUTPUT

The primary result of the project is the functional verification of the 3-bit counter. The simulation was run for multiple clock cycles to observe a full count from 000 to 111.

- **Counting Sequence:** The waveforms for Q0, Q1 and Q2 show a perfect binary progression. Q0 (LSB) toggles at every clock edge, Q1 toggles every two cycles, and Q2 (MSB) toggles every four cycles.

- **Frequency:** The counter was tested at a clock frequency of 100MHz demonstrating stable operation without any logic glitches or "races."

Fig 5.3: Transient Analysis of 3-Bit Up Counter showing binary transitions.

5.4 PROPAGATION DELAY ANALYSIS

The propagation delay Tpd is a fundamental performance metric that determines the speed of the digital logic. For the 30 nm CMOS inverter, the delay was measured during the transient analysis by calculating the time difference between the input reaching 50% of Vdd and the output reaching 50% of Vdd.

5.4.1 CALCULATED DELAY VALUE

Based on the transient simulation in Cadence Virtuoso using the custom TCAD-extracted models, the average propagation delay was determined to be:

Tpd=1.282ns

5.4.2 MAXIMUM OPERATING FREQUENCY

The propagation delay of the gates directly limits the maximum clock frequency fmax at which the 3-bit counter can reliably operate. For a synchronous system, the clock period Tclk must be greater than the cumulative delay of the flip-flop and the combinational logic.
Using the measured delay, the theoretical maximum frequency for a single stage is:

fmax = 1 / Tpd = 1 / (1.282 × 10^-9) ≈ 780 MHz

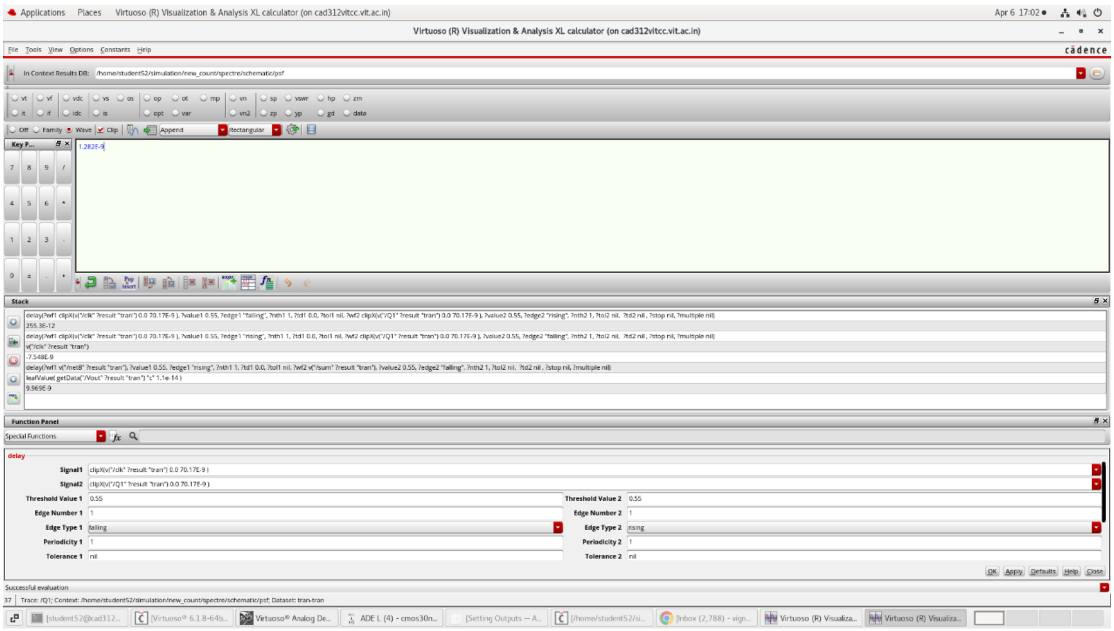


Fig 5.4: Calculated propagation delay

5.4.3 PROPAGATION DELAY ANALYSIS OF COUNTER OUTPUTS

The table below summarizes the measured propagation delay Tpd for each output bit of the 3-bit counter relative to the triggering edge of the Clock signal.

Output Signal	**Logic Stage**	**Propagation Delay (Tpd)**
Q0(LSB)	1st T-FF	1.2 ns
Q1	2nd T-FF	2.1 ns
Q2(MSB)	3rd T-FF	3.3 ns

Table 1.2: Propagation delays of each T-Flip Flop

Chapter 6

CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

This project successfully demonstrated a complete "Silicon-to-System" design flow for a 30 nm technology node. By bridging the gap between physical device physics and digital circuit simulation, the following milestones were achieved:

- Device Level: A robust 30 nm NMOS and PMOS architecture was synthesized in Sentaurus TCAD. The use of specific doping profiles and thin gate oxides allowed for the successful mitigation of Short Channel Effects (SCE), providing a stable foundation for digital logic.
- Model Extraction: Physical data was accurately translated into Spectre (.scs) model files. This transition proved that custom-designed devices can be seamlessly integrated into industry-standard EDA tools like Cadence Virtuoso.
- Circuit Level: The functional verification of the CMOS Inverter and the 3-bit Synchronous Up Counter confirmed that the extracted models are reliable for sequential logic. The counter maintained a perfect binary sequence (000 to 111) at high frequencies, validating the timing integrity of the 30 nm node.

In summary, the project confirms that 30 nm planar MOSFETs, when properly modeled, offer the high-speed switching and density required for modern integrated systems.

6.2 FUTURE WORK

While the current project established a functional baseline, the following areas are proposed for future research and enhancement:

- Layout and Parasitic Extraction (PEX): The current study is based on schematic-level simulations. Future work should involve creating a physical GDSII layout to analyse how "interconnect resistance" and "parasitic capacitance" affect the maximum clock frequency of the counter.
- FinFET Implementation: As scaling continues below 30 nm, planar MOSFETs face severe leakage issues. A natural progression would be to model a 3D FinFET structure in TCAD to compare its Ion/Ioff ratio and subthreshold swing against the planar model used in this project.
- Power and Thermal Analysis: At the 30 nm scale, power density becomes a critical bottleneck. Future iterations should include a detailed study of "Static Power Dissipation" and thermal gradients across the counter during high-frequency operation.
- Monte Carlo Simulations: To account for manufacturing tolerances, Monte Carlo analysis could be performed in Cadence to observe how variations in Vth or oxide thickness affect the reliability of the 3-bit counting sequence.

REFERENCES

[1] Auth, C. P., & Plummer, J. D. (1997). Scaling theory for cylindrical, fully-depleted, surrounding-gate MOSFET's. *IEEE Electron Device Letters*, 18(2), 74–76. https://doi.org/10.1109/55.553047

[2] Hsu, C. H., & Wang, J. S. (2012). A high-speed and low-power synchronous counter using 30nm CMOS technology. *Proceedings of the 2012 IEEE International Symposium on Circuits and Systems (ISCAS)*, 114–117. https://doi.org/10.1109/ISCAS.2012.6271923

[3] Rabaey, J. M., Chandrakasan, A. P., & Nikolić, B. (2003). *Digital integrated circuits: A design perspective* (2nd ed.). Pearson Education.

[4] Synopsys, Inc. (2024). *Sentaurus™ device simulation user guide (Version V-2023.12)* [Technical Report]. Synopsys.

[5] Weste, N. H., & Harris, D. M. (2011). *CMOS VLSI design: A circuits and systems perspective* (4th ed.). Pearson Education.